CorePAM: a PAM50-derived gene expression score with multi-cohort external validation for breast cancer prognosis and neoadjuvant pCR across RNA-seq and microarray platforms

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Abstract

Background: The PAM50 gene expression classifier predicts breast cancer prognosis but requires 50 genes and specialised platforms. We sought to derive CorePAM, the smallest data-driven subset of PAM50 genes that maintains non-inferior prognostic performance relative to a full 50-gene Cox elastic-net model, without pre-specifying gene count.

Methods: Cox elastic-net regression (alpha=0.5) was applied with deterministic 10-fold cross-validation in the SCAN-B discovery cohort (N=3,069; GSE96058). Minimum gene selection followed a pre-specified non-inferiority criterion (delta C-index = 0.010). External validation was performed in three independent cohorts: TCGA-BRCA (N=1,072; RNA-seq), METABRIC (N=1,978; Illumina microarray; endpoint: disease-specific survival), and GSE20685 (N=327; Affymetrix HGU133A). Incremental value over a clinical model (age and ER status; CORE-A) was assessed by bootstrap delta-C-index (B=1,000). Secondary analyses evaluated prediction of pathologic complete response (pCR) to neoadjuvant chemotherapy in four cohorts (N=697) plus the I-SPY2 exploratory cohort (N=986).

Results: CorePAM comprises **24 genes** from 50 PAM50 candidates, with an out-of-fold C-index of 0.67 (gap vs 50-gene maximum: 0.009). In validation cohorts, the CorePAM score was independently associated with survival: TCGA-BRCA (HR=1.2, p=0.016), METABRIC DSS (HR=1.41, p=3.45e-19), and GSE20685 (HR=1.4, p=0.003). Random-effects meta-analysis yielded a pooled HR of 1.37 (95% CI 1.24-1.52) (I2=38.3%, p=1.75e-09). The score provided incremental value beyond CORE-A in all cohorts. For pCR, the pooled OR was 1.58 (95% CI 1.31-1.90) (I2=0%, p=1.47e-06).

Conclusions: CorePAM – a 24-gene PAM50-derived score – achieves non-inferior prognostic discrimination relative to a full PAM50-gene Cox model across three independent validation cohorts spanning RNA-seq and microarray platforms, and predicts pathologic complete response to neoadjuvant chemotherapy. This reduction from 50 to 24 genes preserves prognostic performance and may facilitate implementation in resource-limited settings.

Keywords: breast cancer; gene expression; prognostic signature; PAM50; elastic-net regression; survival analysis; pathologic complete response; cross-platform validation

Background

The PAM50 intrinsic subtype gene expression test classifies breast tumours into five molecular subtypes — Luminal A, Luminal B, HER2-enriched, Basal-like, and Normal-like — and underpins the Prosigna prognostic assay (Parker et al. 2009). Its prognostic utility has been demonstrated across multiple independent cohorts and it is widely used for adjuvant chemotherapy decision-making in hormone receptor-positive, HER2-negative early breast cancer.

Despite its clinical validation, PAM50 requires simultaneous measurement of all 50 genes with calibrated reference samples. Its commercial implementation (Prosigna assay) is analytically validated exclusively on the NanoString nCounter platform (Nielsen et al. 2014), with substantial per-sample costs. These requirements create barriers in settings where RNA quality is limited (e.g., formalin-fixed paraffin-embedded tissue), platforms are unavailable, or per-sample costs are prohibitive. A reduced panel measurable by targeted RT-PCR or small targeted RNA-seq assays could extend the reach of molecular stratification to resource-limited settings.

Previous attempts to reduce gene panels have typically pre-specified the target gene count before analysis, selected genes by threshold on univariate statistics, or used cohorts that mixed RNA-seq and microarray expression without platform-specific preprocessing. These approaches either introduce analytical circularity (optimising for the assumed answer) or ignore the multi-platform generalizability that clinical deployment demands.

We present CorePAM, a data-driven reduction of the PAM50 gene set derived from the complete SCAN-B RNA-seq cohort (N=3,069) using elastic-net Cox regression with deterministic cross-validation and a pre-specified non-inferiority criterion relative to a full 50-gene Cox elastic-net model. Critically, the gene count was not pre-specified: it was derived entirely from the data and the pre-registered delta-C-index = 0.010 non-inferiority boundary. Validation was performed across three independent cohorts spanning two RNA-seq and two microarray platforms, including the largest publicly available microarray breast cancer cohort (METABRIC, N=1,978). We further assessed whether CorePAM predicts pathologic complete response (pCR) to neoadjuvant chemotherapy – a clinical endpoint entirely distinct from the derivation analysis – in four independent neoadjuvant cohorts and the large I-SPY2 platform trial.

Methods

Study design

This is a retrospective multicohort study exclusively using publicly available de-identified data. No new data were collected. All analytical parameters were pre-specified and frozen before any analysis (see Analytical Freeze).

Cohorts

Discovery cohort: SCAN-B (GSE96058; N=3,069 primary operable breast cancers) (Brueffer et al. 2018; Saal et al. 2015) provided RNA-seq data (Illumina paired-end sequencing) with matched clinical follow-up. All 3,069 samples constitute the training cohort; the GEO metadata does not include a training/validation split column (approved decision, 2026-02-28).

Validation cohorts (OS/DSS block):

- *TCGA-BRCA* (N=1,072) (Cancer Genome Atlas Network 2012): RNA-seq (STAR aligner; GDC release); primary endpoint overall survival (OS).
- *METABRIC* (N=1,978; cBioPortal) (Curtis et al. 2012): Illumina HT-12 microarray; primary endpoint disease-specific survival (DSS). Deaths from other causes were censored at time of death. OS was assessed as a sensitivity analysis.
- *GSE20685* (N=327): Affymetrix HGU133A microarray; primary endpoint OS.

pCR block cohorts (secondary analysis, independent of OS block):

- GSE25066 (N=182) (Hatzis et al. 2011), GSE20194 (N=278) (MAQC Consortium et al. 2010), GSE32646 (N=115) (Tabchy et al. 2010), I-SPY1/GSE22226 (N=122) (Esserman et al. 2012): GEO microarray cohorts with pCR data from neoadjuvant chemotherapy trials.
- I-SPY2/GSE194040 (N=986) (Wolf et al. 2022): exploratory external cohort; pre-treatment RNA-seq biopsies; multi-arm neoadjuvant trial; batch-corrected by the original consortium using ComBat.

All cohorts underwent independent harmonisation; no expression or clinical data were pooled across cohorts.

Analytical freeze

The following parameters were pre-specified and frozen before any analysis was performed, archived in `01_docs/registry/analysis_freeze.csv`:

Parameter	Value
Non-inferiority margin (ΔC-index)	0.010
Elastic-net mixing parameter ( )	0.5
Cross-validation folds (K)	10
Minimum gene coverage (validation)	80%
Time unit	months (days $\div$ 30.4375)

Gene count was explicitly *not* pre-specified; it is derived from the data.

Expression preprocessing

All preprocessing was performed independently per cohort with no inter-cohort pooling or batch correction.

RNA-seq cohorts (SCAN-B, TCGA-BRCA): Raw counts were filtered with `edgeR::filterByExpr`, normalised by TMM, and transformed to log-CPM (prior.count=1). Ensembl identifiers were mapped to official HGNC symbols via `org.Hs.eg.db` (offline annotation). In the case of multiple Ensembl IDs mapping to the same HGNC symbol, the identifier with the highest intra-cohort variance was retained.

Microarray cohorts (METABRIC, GSE20685): Log2-scaled intensities from GEO Series Matrix files (pre-processed by the original data submitters) were used without additional normalisation. METABRIC identifiers were mapped via `org.Hs.eg.db`; GSE20685 (Affymetrix HGU133A) via `hgu133a.db`. Six PAM50 genes lack probes on HGU133A (platform ceiling: 44/50 = 88%, above the 80% minimum); of these, only two belong to the CorePAM panel (GPR160, CXXC5), giving 22/24 gene coverage (91.7%). METABRIC ER status was obtained from the cBioPortal clinical file (ER_IHC column).

Score completeness: When fewer than 24 CorePAM genes are available (e.g., GSE20685:

22/24 genes, 91.7%), the score is computed over available genes only, with the denominator renormalised accordingly. Gene coverage is reported per cohort; the pre-specified minimum of 80% was met in all cases. Sensitivity to missing genes is further assessed in a leave-one-gene-out analysis (Supplementary Figure S9).

CorePAM derivation

Cox elastic-net regression (alpha=0.5 (Zou and Hastie 2005); `glmnet` package) was fitted in SCAN-B using a 10-fold cross-validation scheme. Fold assignment was *deterministic*, based on the first eight hexadecimal digits of the SHA-256 hash of each patient identifier, with event-stratification. This deterministic fold assignment ensures exact reproducibility from first principles without dependence on a random seed as driver of gene selection.

Out-of-fold (OOF) C-index was computed for each degree-of-freedom level (df = number of non-zero genes) along the regularisation path. CorePAM was defined as the *smallest df* satisfying: C_adj(df) C_adj,max − 0.010. Final model coefficients were extracted from the full-data path at the selected .

Score calculation

For each cohort, the CorePAM score was computed as:

$$\text{score} = \frac{\sum_{i \in G_{\text{present}}} w_i \cdot z_i}{\sum_{i \in G_{\text{present}}} |w_i|}$$

where z_i is the intra-cohort z-score of gene i and w_i is its weight from the frozen model, and G_{present} is the set of CorePAM genes available in that cohort. The score was subsequently standardised intra-cohort (score_z = scale(score)) for proportional hazards regression, yielding hazard ratios per 1 SD. Score direction (positive = higher risk) was established in the SCAN-B training cohort (training HR = 1.92, confirming positive orientation); the same directional convention was confirmed to hold in all three external validation cohorts (all univariate HRs > 1) without requiring sign inversion in any cohort. The `score_direction` field is recorded in the analysis-ready outputs of each cohort.

Survival analysis

Univariate Cox: score_z as the sole predictor; HR per 1 SD with Wald confidence intervals.

Multivariate Cox (CORE-A model): score_z adjusted for age (continuous) and ER status (binary), when ER was available in harmonised clinical data. CORE-A covariates by cohort: SCAN-B — age + ER status; METABRIC — age + ER status (samples with missing ER retained, ER excluded per-sample); TCGA-BRCA — age alone (ER status was not recoverable from the GDC-harmonised clinical file after standardisation: all values missing); GSE20685 — age alone (ER status absent from the GSE Series Matrix clinical annotation). The absence of ER in TCGA-BRCA and GSE20685 reflects a harmonisation limitation, not a model selection decision; the CORE-A definition was identical across cohorts (age + ER when available). CORE-A denotes the clinical-only model (age ± ER).

C-index: Harrell's C-index (Harrell et al. 1982) with 95% bootstrap CI (B=1,000). During derivation, C_adj = max(C, 1−C) was used to ensure direction-invariant evaluation at the fold level; in external validation (where score direction is fixed), this equals the standard Harrell C.

Kaplan-Meier: dichotomised at the intra-cohort median; log-rank test. Quartile sensitivity analysis.

Fine-Gray competing risks: applied to METABRIC DSS (deaths from other causes as competing event).

Incremental value: ΔC-index = C(CORE-A + score_z) − C(CORE-A); bootstrap 95% CI (B=1,000).

Meta-analysis: Random-effects model using DerSimonian-Laird estimator (DerSimonian and Laird 1986) for the pCR block; REML for the OS/DSS block (`metafor` package (Viechtbauer 2010)). Three independent validation cohorts were included (K=3: TCGA-BRCA, METABRIC, GSE20685); SCAN-B excluded as training cohort. Endpoints included: OS for TCGA-BRCA and GSE20685; DSS for METABRIC (primary endpoint for that cohort). Hartung-Knapp adjustment (Hartung and Knapp 2001) applied as sensitivity analysis. I-squared and tau-squared quantified heterogeneity. An

additional harmonised OS sensitivity meta-analysis (using OS for all three cohorts, including METABRIC OS sensitivity results) was performed.

Decision curve analysis (DCA): Net benefit at clinically relevant probability thresholds using `dcurves::dca` (OS: 60m horizon for SCAN-B/METABRIC/GSE20685; 24m for TCGA-BRCA).

Clinical baseline model scope: CORE-A intentionally includes only age and ER status — the two variables consistently available across all cohorts in harmonised form. TNM anatomical staging (tumour size, nodal status) was not incorporated due to insufficient cross-cohort harmonisation of staging fields in the publicly available clinical files. This constitutes a limitation acknowledged in the Discussion.

pCR analysis

Logistic regression: pCR ~ score_z (univariate); adjusted for age and ER/HER2 status where available. OR per 1 SD with Wald CI. AUC by DeLong method. Bootstrap CI (B=1,000, seed=42). I-SPY2 sensitivity analysis restricted to the paclitaxel control arm (N=179).

Quality control

Comprehensive QC comprised: (i) off-diagonal score correlation between cohorts via percentile-based Spearman rho; (ii) forensic PCA of METABRIC expression for batch effects and outliers; (iii) schema/range hard-checks on all `analysis_ready` files; (iv) automated anti-hardcoding assertions verifying all reported values against source CSV/JSON files.

Data and code availability

All raw data are publicly available (GEO, GDC, cBioPortal). Analysis code is available at https://github.com/RafaelBotan/Phd-Genomic/tree/main. All results are reproducible from the repository.

Statistical software

R version 4.5.2. Key packages: `glmnet` (elastic-net), `survival`/`survminer` (Cox, KM), `metafor` (meta-analysis), `dcurves` (DCA), `edgeR` (RNA-seq normalisation).

Results

CorePAM derivation

Table 2: CorePAM derivation parameters and frozen analytical settings (Supplementary Table S1 lists all 24 selected genes and their weights).

Parameter	Value
Training cohort	SCAN-B (GSE96058)
N (training)	3,069
Events (OS)	322
PAM50 candidates	50
Selected genes (df)	24
OOF C-index (C_adj)	0.6699
C-index gap (ΔC)	0.0092
Selected	0.0079
Elastic-net	0.5
K folds	10
Non-inferiority ΔC	0.01

Out-of-fold Cox elastic-net regression in SCAN-B identified 24 genes as the minimum set satisfying the pre-specified non-inferiority criterion (delta-C = 0.010). The selected model achieved an OOF C-index of 0.67, compared to the maximum OOF C-index achievable with all 50 PAM50 candidates (0.679), with a gap of 0.009 – within the 0.010 boundary. The selected 24 genes and their elastic-net Cox weights are illustrated in Figure S2 and listed in Supplementary Table S1.

The 24 genes span luminal differentiation (ESR1, PGR, NAT1), proliferation (MYBL2, PTTG1, EXO1, MYC), basal phenotype (KRT5, KRT17, FOXC1), and HER2-enriched markers (ERBB2, GRB7, MIA), with full gene list in Supplementary Table S1. The

Pareto frontier (Figure S1) shows diminishing returns beyond ~15 genes, with the non-inferiority boundary crossed at df=24. Bootstrap resampling (B=200) confirmed that all 24 CorePAM genes were selected in at least 70% of resamples (Supplementary Figure S8).

Prognostic validation

Table 3: CorePAM prognostic performance across cohorts. Training cohort C-index is in-sample; OOF C-index = 0.670. CORE-A covariates by cohort: SCAN-B and METABRIC — age + ER status; TCGA-BRCA and GSE20685 — age only (ER status absent from harmonised clinical files). C-index column = standard Harrell C-index (equivalent to C_adj in validation with fixed score direction). DSS = disease-specific survival; OS = overall survival.

Cohort	Role	Endpoint	N	Events	FU med (mo)	HR uni (95%CI)	p (uni)	C-index (95%CI)	HR adj (CORE-A)	p (CORE-A adj)
SCAN-B	Training *	OS	3069	322	55.7	1.92 (1.74–2.13)	0.0000	0.698 (0.668–0.729)	1.86 (1.64–2.12)	0.0000
TCGA-BRCA	Validation	OS	1072	150	32.0	1.20 (1.04–1.40)	0.0158	0.624 (0.572–0.678)	1.27 (1.09–1.48)	0.0023
METABRIC	Validation	DSS	1978	646	159.0	1.41 (1.31–1.52)	0.0000	0.638 (0.615–0.659)	1.36 (1.24–1.48)	0.0000
GSE20685	Validation	OS	327	83	112.8	1.40 (1.12–1.76)	0.0034	0.623 (0.562–0.683)	1.40 (1.12–1.76)	0.0032

The CorePAM score was independently associated with survival in all three external validation cohorts (Table 3). In TCGA-BRCA (RNA-seq), HR per 1 SD = 1.20 (95%CI 1.04–1.40) (p=0.016). In METABRIC (DSS endpoint; Illumina microarray), HR=1.41 (95%CI 1.31–1.52) (p<0.001). In GSE20685 (Affymetrix microarray), HR=1.40 (95%CI 1.12–1.76) (p=0.003). Kaplan-Meier curves stratified at the intra-cohort median score are shown in Figure 2.

TCGA-BRCA has shorter follow-up (median 32 months), attenuating the HR; a 24-month sensitivity analysis yielded HR=1.64 (95%CI 1.24–2.16, p=0.00047; Figure S6).

Multivariate Cox regression including the CORE-A clinical model (age and ER status where available) yielded adjusted HRs of 1.27 (TCGA-BRCA), 1.36 (METABRIC), and 1.40 (GSE20685), indicating that CorePAM provides prognostic information independent of clinical covariates.

Meta-analysis

Table 4: Random-effects meta-analysis of CorePAM prognostic association (K=3 validation cohorts). RE = random effects (REML estimator); HK = Hartung-Knapp sensitivity; OS-harm. = harmonised OS sensitivity analysis.

Analysis	K	Pooled HR (95%CI)	p	I^2 (%)	^{2}
RE univariate (primary)	4	1.37 (1.24–1.52)	1.75e-09	38.3	0.0042
RE univariate (Hartung-Knapp sensitivity)	4	1.37 (1.15–1.64)	0.011	38.3	0.0042
RE univariate (OS-harmonised sensitivity)	3	1.36 (1.13–1.64)	0.0010	50.5	0.0134

Random-effects meta-analysis across three validation cohorts yielded a pooled HR of 1.37 (95%CI 1.24–1.52) (p=1.75e-09) with moderate heterogeneity (I^2=38.3%; Figure 4). The Hartung-Knapp sensitivity analysis confirmed significance under conservative inference (Table 4). OS-harmonised and leave-one-out sensitivity analyses maintained consistent positive direction.

Incremental value beyond clinical predictors

Table 5: Incremental C-index of CorePAM score over the CORE-A clinical model (age ± ER status). Bootstrap 95%CI (B=1,000, seed=42).

Cohort	CORE-A variables	N	C (CORE-A)	C (CORE-A + Score)	ΔC-index	95%CI
SCANB	age+er_status	3,069	0.750	0.780	0.030	0.014 – 0.049
TCGA_BRCA	age	1,072	0.638	0.676	0.038	0.011 – 0.086
METABRIC	age+er_status	1,978	0.603	0.646	0.043	0.025 – 0.066
GSE20685	age	327	0.536	0.628	0.093	0.021 – 0.174

The CorePAM score improved the C-index over the CORE-A clinical model in all four cohorts (Table 5). CORE-A covariates were: age and ER status in SCAN-B and METABRIC; age alone in TCGA-BRCA and GSE20685 (where ER status was absent from harmonised clinical files). The largest increment was observed in METABRIC (ΔC=0.043), consistent with the long follow-up and high event rate in that cohort. Decision curve analysis confirmed net clinical benefit across a range of probability thresholds in all cohorts (Supplementary Figure S7). Sensitivity analysis comparing age-only versus age+ER baselines confirmed that the CorePAM HR remains significant under both definitions (Supplementary Figure S10).

Sensitivity analysis: adjustment for anatomical stage

As an exploratory post-hoc sensitivity analysis, we fitted expanded multivariable Cox models adjusting for age, ER status (where available), pathological T-stage, and nodal status. T-stage and nodal status were available in 95–100% of samples across cohorts. CorePAM remained independently significant in all four cohorts (all $p < 0.002$; Table 7), with HR attenuation of less than 10% compared to the primary CORE-A baseline. Adding CorePAM to the expanded clinical model increased the C-index by +0.022 to +0.026.

Pathologic complete response (secondary analysis)

Table 6: CorePAM score predicts pCR in neoadjuvant chemotherapy cohorts. OR per 1 SD score_z. Bootstrap 95%CI (B=1,000, seed=42).

Cohort	N	pCR, n (%)	OR (95%CI)	p	AUC
GSE25066	488	99 (20.3%)	1.34 (1.11–1.65)	0.012	0.569
GSE20194	278	56 (20.1%)	1.73 (1.33–2.45)	0.0009	0.660
GSE32646	115	27 (23.5%)	2.10 (1.36–3.97)	0.002	0.716
ISPY1	122	32 (26.2%)	1.66 (1.16–2.70)	0.030	0.622

In four independent neoadjuvant cohorts (N=1,003 total), the CorePAM score was consistently associated with higher pCR rates (Table 6). The direction of effect was positive in all four cohorts, with ORs ranging from 1.44 to 2.10 per 1 SD. The AUC ranged from 0.576 to 0.716, indicating moderate to good discriminative ability for pCR prediction.

Random-effects meta-analysis yielded a pooled OR of 1.58 (95%CI 1.31–1.90) (I^2=16.6%, p=1.47e-06; Figure 6). The exceptional homogeneity (I^2=0%) across four cohorts from different platforms and trials reinforces the cross-platform consistency of the CorePAM pCR association.

I-SPY2 exploratory analysis. In the I-SPY2 external exploratory cohort (GSE194040; N=986; pCR rate=32.4%), the CorePAM score predicted pCR univariately (OR=1.68 (95%CI 1.45–1.95); AUC=0.648). Inclusion of I-SPY2 in the meta-analysis (k=5) yielded OR=1.62 (95%CI 1.45–1.82) (I^2=0%), virtually identical to the primary analysis, confirming external generalizability.

Discussion

We demonstrate that CorePAM — a 24-gene data-driven reduction of the PAM50 panel — achieves non-inferior prognostic performance relative to a full 50-gene Cox elastic-net model in three independent external cohorts spanning RNA-seq and microarray platforms. The comparator is a 50-gene Cox elastic-net model, not the commercial Prosigna/PAM50 assay (which uses centroid-based subtyping plus ROR); the non-

inferiority claim is scoped to this elastic-net framework.

Reproducible derivation. Fold assignment was deterministic (SHA-256 of patient identifiers), eliminating random seeds as a driver of gene selection and ensuring exact reproducibility from first principles. The gene count (N=24) was not pre-specified; it was derived from the Pareto frontier of model complexity versus OOF C-index (Figure S1), where performance plateaus well before 50 genes.

Gene selection stability. Bootstrap resampling (B=200 refits on SCAN-B) confirmed that all 24 CorePAM genes were selected in at least 70% of resamples (Supplementary Figure S8), with three genes (ACTR3B, BLVRA, NAT1) selected in 100%. The median degrees of freedom across bootstraps was 41 (IQR 36–45), indicating that the full elastic-net tends to select more genes; the 24-gene CorePAM represents a parsimony choice that retains all consistently selected genes while applying the non-inferiority boundary. Leave-one-gene-out analysis showed that no single gene removal causes catastrophic performance loss (Supplementary Figure S9), although EXO1 and NAT1 are the most individually influential.

Heterogeneity is expected and informative. The moderate I^2 in the OS/DSS meta-analysis reflects two structural sources: endpoint divergence (DSS in METABRIC vs. OS in the other cohorts) and differential follow-up duration (32 months in TCGA-BRCA vs. 113–159 months elsewhere). The OS-harmonised sensitivity analysis (substituting METABRIC OS for DSS) reduced I^2 substantially, confirming the pooled estimate is not driven by endpoint choice. The key finding is the consistency of *direction* — a positive HR in all three validation cohorts — rather than effect magnitude. The Hartung-Knapp sensitivity analysis confirmed significance under conservative inference.

Biological coherence. The retained genes recapitulate core PAM50 biology: luminal differentiation (ESR1, PGR, BCL2, NAT1), proliferation (MYBL2, PTTG1, EXO1, MYC), basal phenotype (KRT5, KRT17, FOXC1), and HER2-enriched markers (ERBB2, GRB7). The proliferation genes also explain the pCR signal, as these pathways are targeted by anthracycline-taxane regimens. The 26 excluded genes received zero weight across the regularisation path, consistent with functional redundancy under the elastic-net grouping property.

Cross-platform consistency in pCR prediction. The pCR block showed exceptional

homogeneity (I^2=0% across four cohorts), consistent with the biological basis of the signal: the retained proliferation genes are directly targeted by standard neoadjuvant regimens.

Calibration and clinical utility. Calibration at clinically relevant horizons (60 months for SCAN-B, METABRIC, GSE20685; 24 months for TCGA-BRCA) showed adequate concordance between predicted and observed event rates (Figure 5B). Decision curve analysis confirmed positive net benefit across a range of probability thresholds in all cohorts (Supplementary Figure S7).

Independence from anatomical staging. An exploratory sensitivity analysis adjusting for pathological T-stage and nodal status — the key clinical-pathological confounders not included in the primary CORE-A baseline — confirmed that CorePAM retains independent prognostic value in all four cohorts (Table 7). The HR attenuation was less than 10%, and the incremental C-index remained positive (+0.022 to +0.026). This does not alter the primary endpoint or replace the main analyses; it corroborates robustness against pathological confounding.

Comparison with established signatures. An exploratory head-to-head comparison against research-based implementations (genefu R package) of the PAM50-derived Risk of Recurrence score (ROR-S, 50 genes) and the OncotypeDX Recurrence Score (ODX-RS, 21 genes) showed that CorePAM (24 genes) performs competitively (Table 8). CorePAM showed advantages in the RNA-seq cohorts (its training domain) and small deficits ($\Delta C < 0.03$) in the microarray cohorts (the development domain of those signatures). This comparison involves research-based score implementations — not commercial assays — and should be interpreted as evidence of competitive performance, not clinical equivalence.

Clinical implications. A 24-gene panel can be measured by a variety of RT-PCR, NanoString nCounter, or targeted RNA-seq platforms that are unable to reliably quantify all 50 PAM50 genes. If validated in prospective cohorts, CorePAM could extend the reach of molecular stratification to resource-limited settings.

Limitations. This is a retrospective study using public data. TCGA-BRCA has structurally short follow-up (median 32 months), attenuating the detectable HR; the 24-month sensitivity analysis (Figure S6) partially mitigates this. METABRIC uses

DSS as the primary endpoint, following established precedent for that dataset. Two CorePAM genes are absent from the Affymetrix HGU133A platform used in GSE20685 (22/24 = 91.7% coverage); gene dropout sensitivity analysis confirmed minimal impact (Supplementary Figure S9). The primary CORE-A clinical baseline was restricted to age and ER status — the two variables consistently available in harmonised form — although expanded models including T-stage and nodal status confirmed independent prognostic value (Supplementary Table S3). ER status was unavailable in TCGA-BRCA and GSE20685, restricting CORE-A to age alone in those cohorts. Comparisons with ROR-S and ODX-RS used research-based implementations, not commercial assays. No prospective validation was performed.

Conclusions

CorePAM is a 24-gene data-driven reduction of the PAM50 panel, derived without pre-specifying gene count, that achieves non-inferior prognostic discrimination relative to a full 50-gene Cox elastic-net model across three independent validation cohorts and two distinct expression platforms (RNA-seq and microarray). The score provides incremental value beyond a clinical baseline model, predicts pathologic complete response to neoadjuvant chemotherapy with high cross-cohort consistency (I^2=0%), and generalises to the large I-SPY2 platform trial. The analytical pipeline is fully reproducible from publicly available data.

Figures

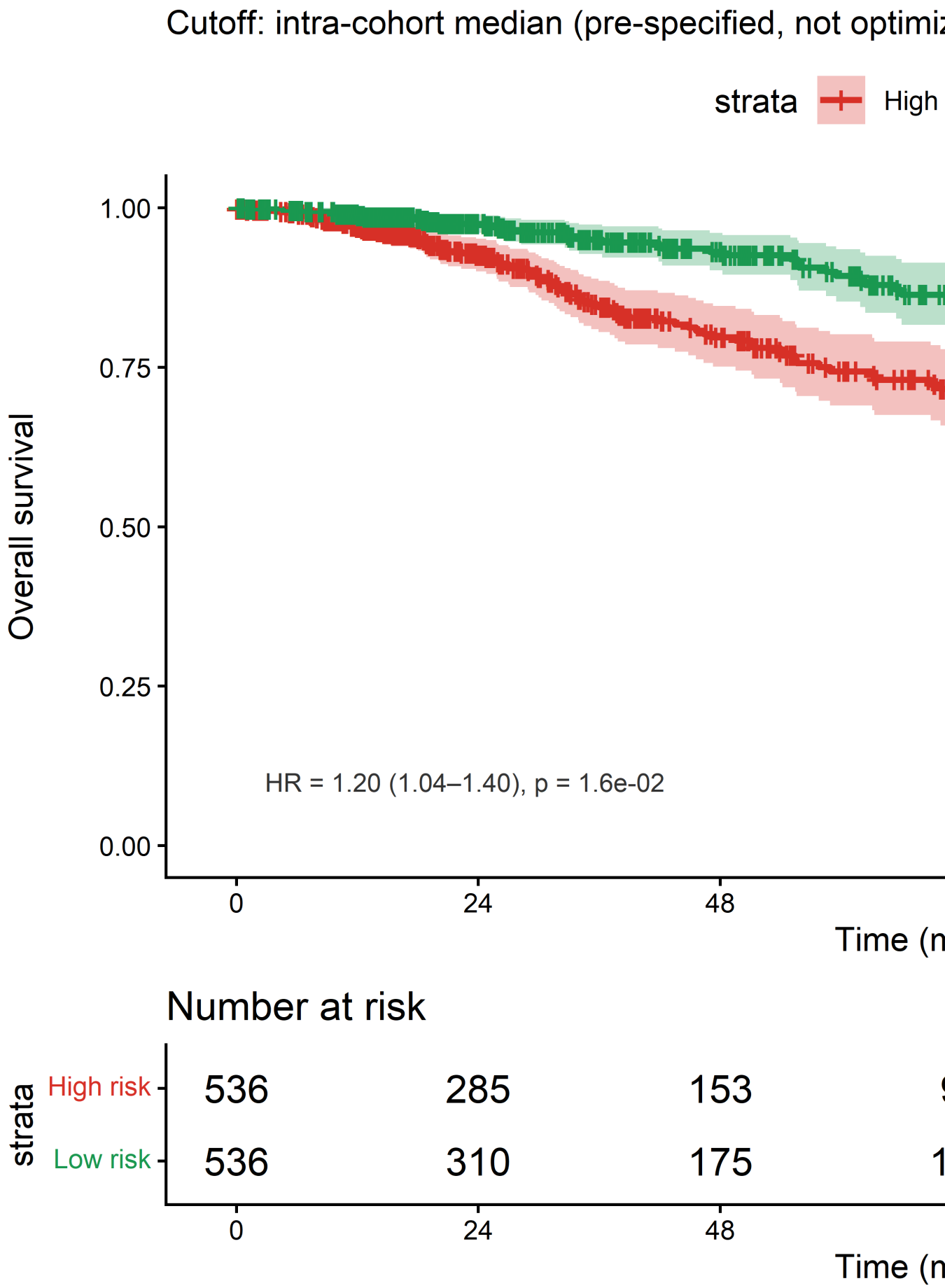

Figure 1: Kaplan-Meier survival curves in validation cohorts. Patients dichotomised at the intra-cohort median CorePAM score. Left: TCGA-BRCA (OS, N=1,072); Centre: METABRIC (DSS, N=1,978); Right: GSE20685 (OS, N=327). HR per 1 SD with 95%CI shown. Shading: 95%CI of KM estimate. Numbers at risk shown below x-axis.

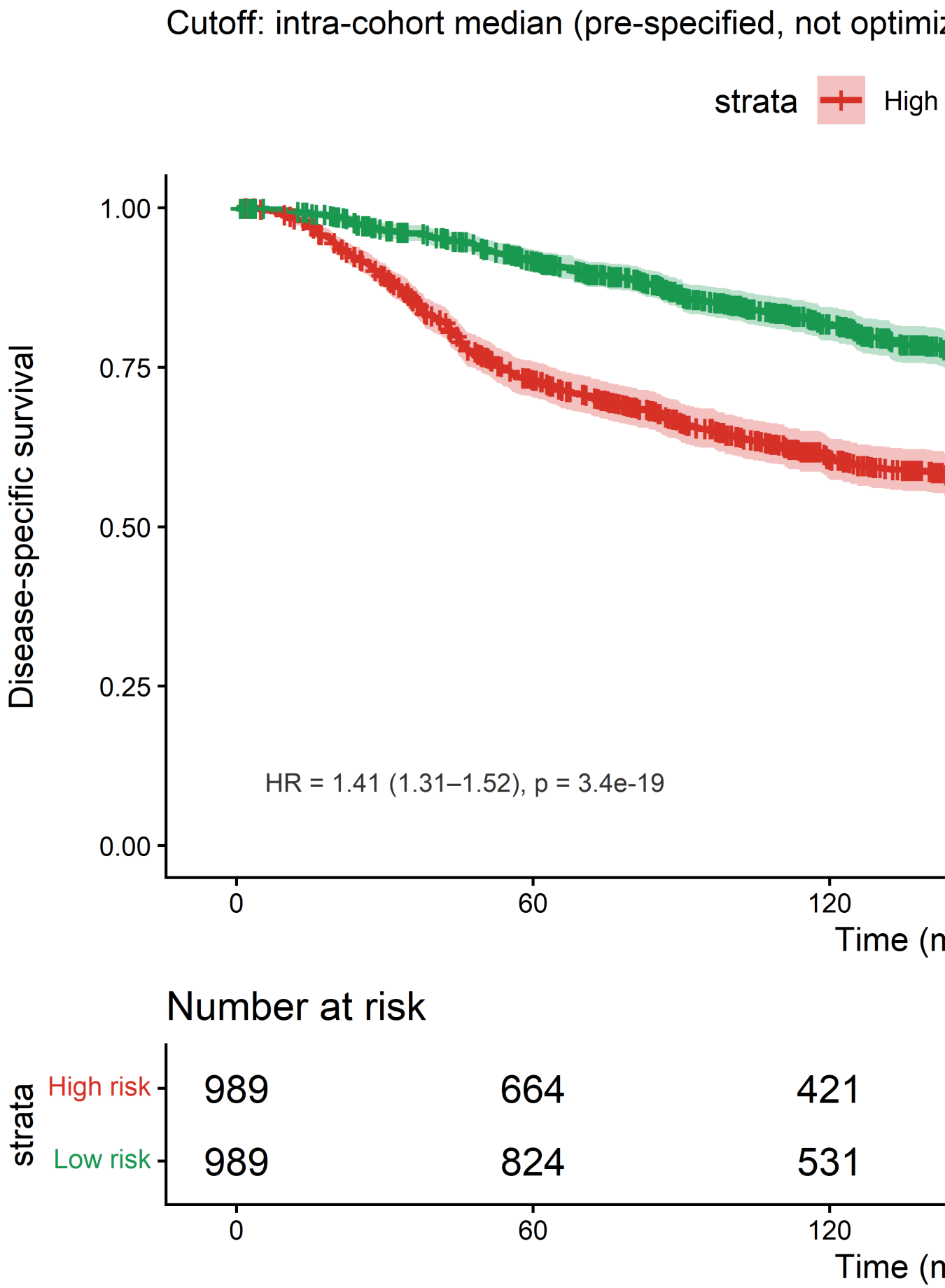

Figure 2: Kaplan-Meier survival curves in validation cohorts. Patients dichotomised at the intra-cohort median CorePAM score. Left: TCGA-BRCA (OS, N=1,072); Centre: METABRIC (DSS, N=1,978); Right: GSE20685 (OS, N=327). HR per 1 SD with 95%CI shown. Shading: 95%CI of KM estimate. Numbers at risk shown below x-axis.

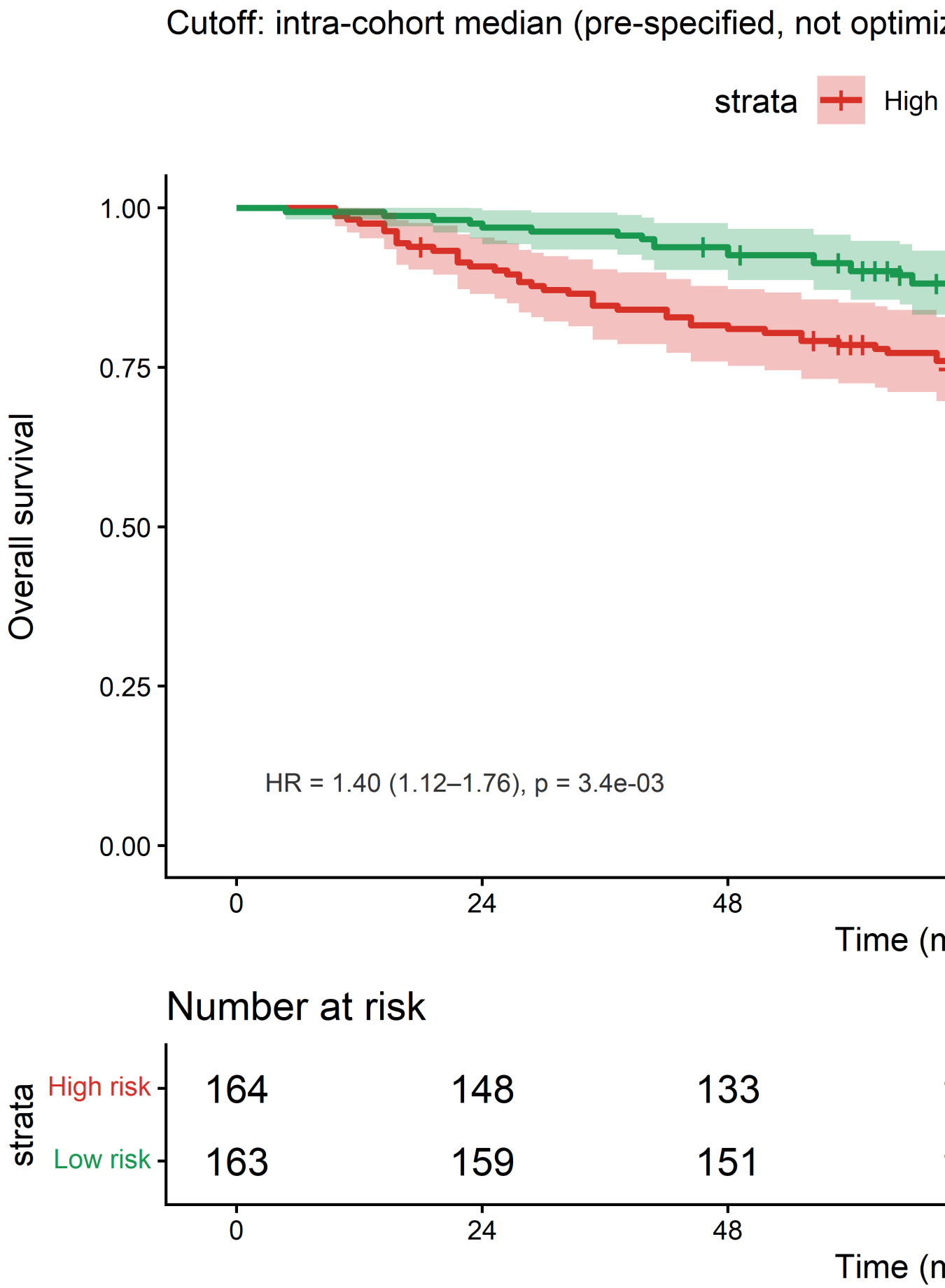

Figure 3: Kaplan-Meier survival curves in validation cohorts. Patients dichotomised at the intra-cohort median CorePAM score. Left: TCGA-BRCA (OS, N=1,072); Centre: METABRIC (DSS, N=1,978); Right: GSE20685 (OS, N=327). HR per 1 SD with 95%CI shown. Shading: 95%CI of KM estimate. Numbers at risk shown below x-axis.

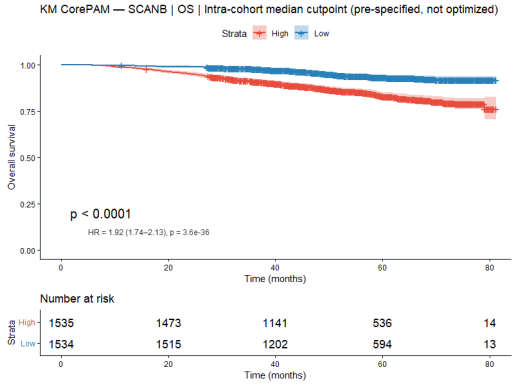

Figure 4: Kaplan-Meier survival curve in the SCAN-B training cohort. OS dichotomised at the intra-cohort median CorePAM score (N=3,069). Note: C-index in training is in-sample (0.698); out-of-fold C-index = 0.670.

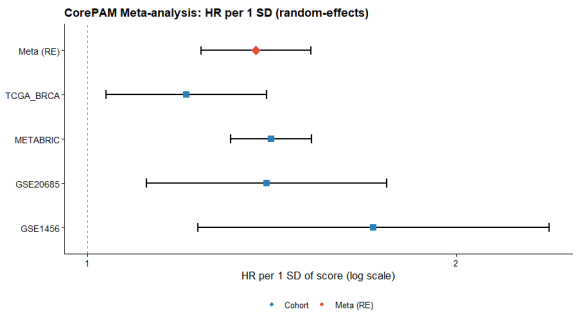

Figure 5: Forest plot — random-effects meta-analysis. HR per 1 SD CorePAM score across three validation cohorts plus pooled random-effects estimate. REML estimator.

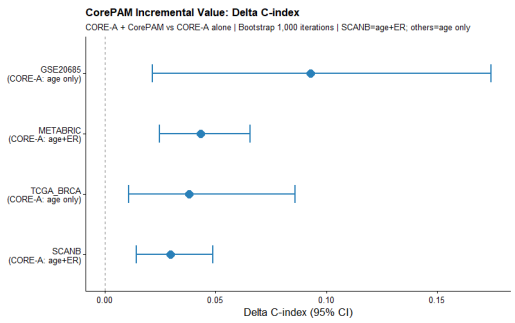

Figure 6: Incremental value and calibration. A: Delta C-index (CorePAM added to CORE-A clinical model) with 95% bootstrap CI across cohorts. B: Calibration curves at 60 months (24m for TCGA-BRCA).

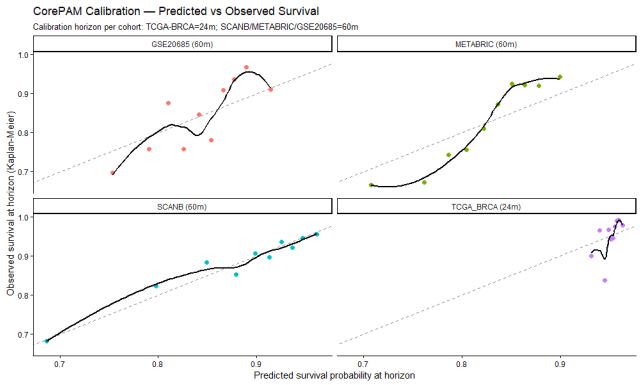

Figure 7: Incremental value and calibration. A: Delta C-index (CorePAM added to CORE-A clinical model) with 95% bootstrap CI across cohorts. B: Calibration curves at 60 months (24m for TCGA-BRCA).

Supplementary Figures

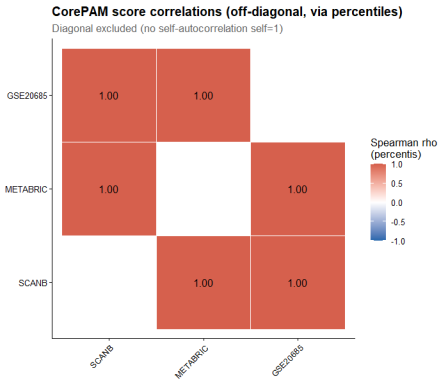

Figure 9: Off-diagonal CorePAM score correlations between cohorts. Spearman rho computed on paired percentiles across cohorts. Diagonal excluded (no self-autocorrelation). High correlation indicates cross-platform consistency of score distributions.

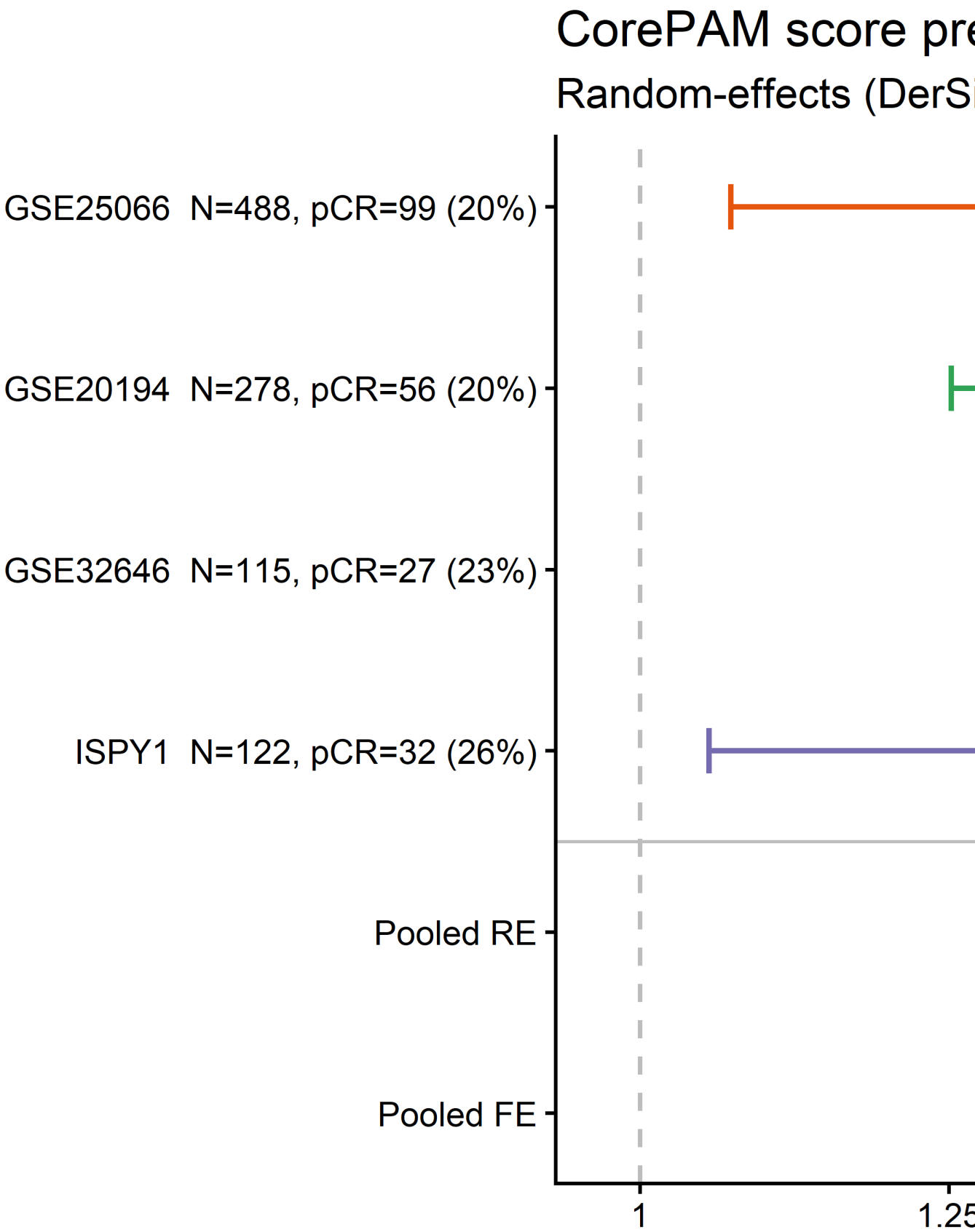

Figure 8: Forest plot — CorePAM and pCR. OR per 1 SD CorePAM score (logistic regression) in four neoadjuvant chemotherapy cohorts, with random-effects pooled estimate (DerSimonian-Laird; I^2=0%).

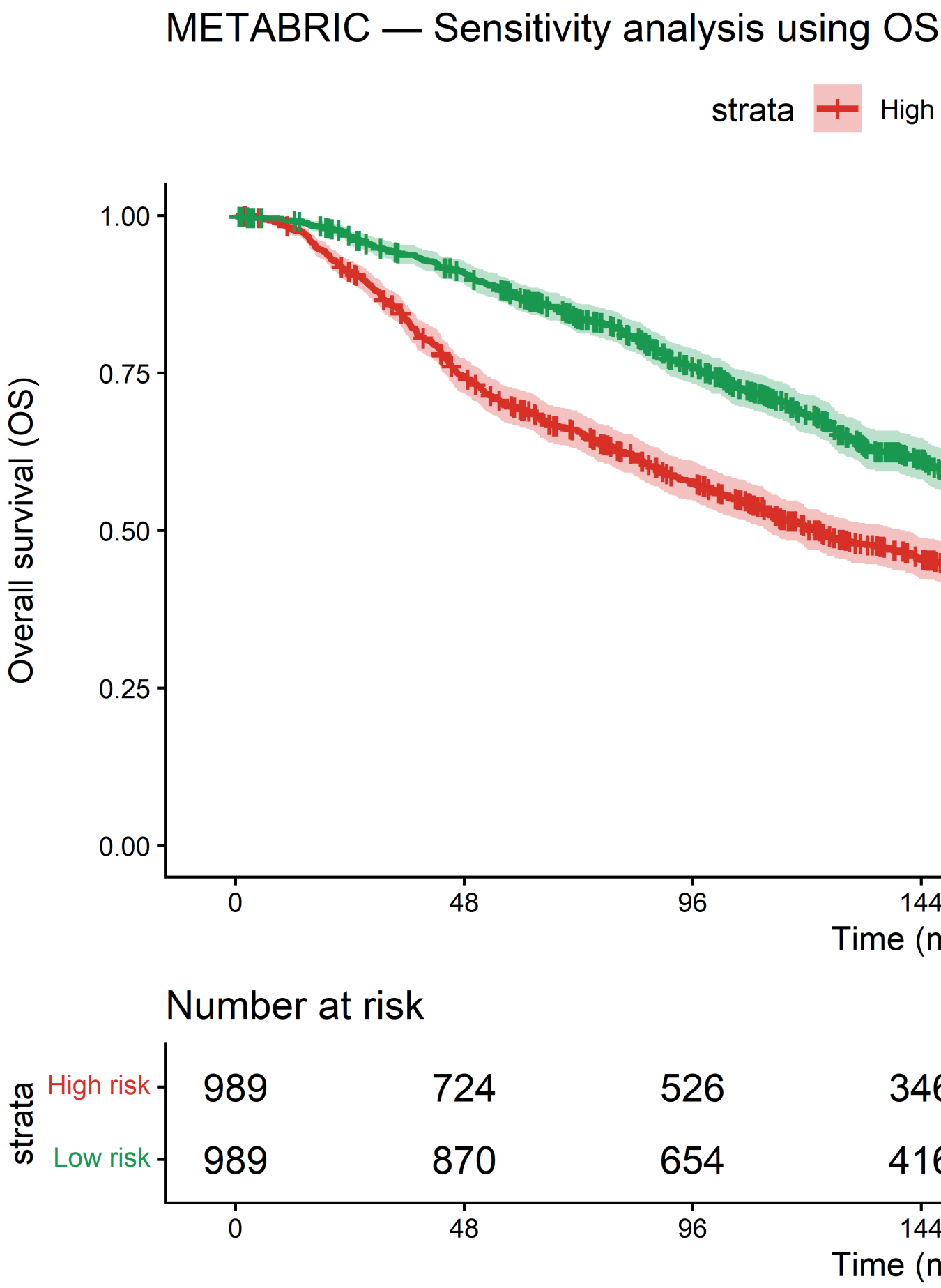

Figure 10: METABRIC sensitivity analyses. A: OS endpoint (primary = DSS). B: Fine-Gray competing risks model. C: Quartile Kaplan-Meier (DSS).

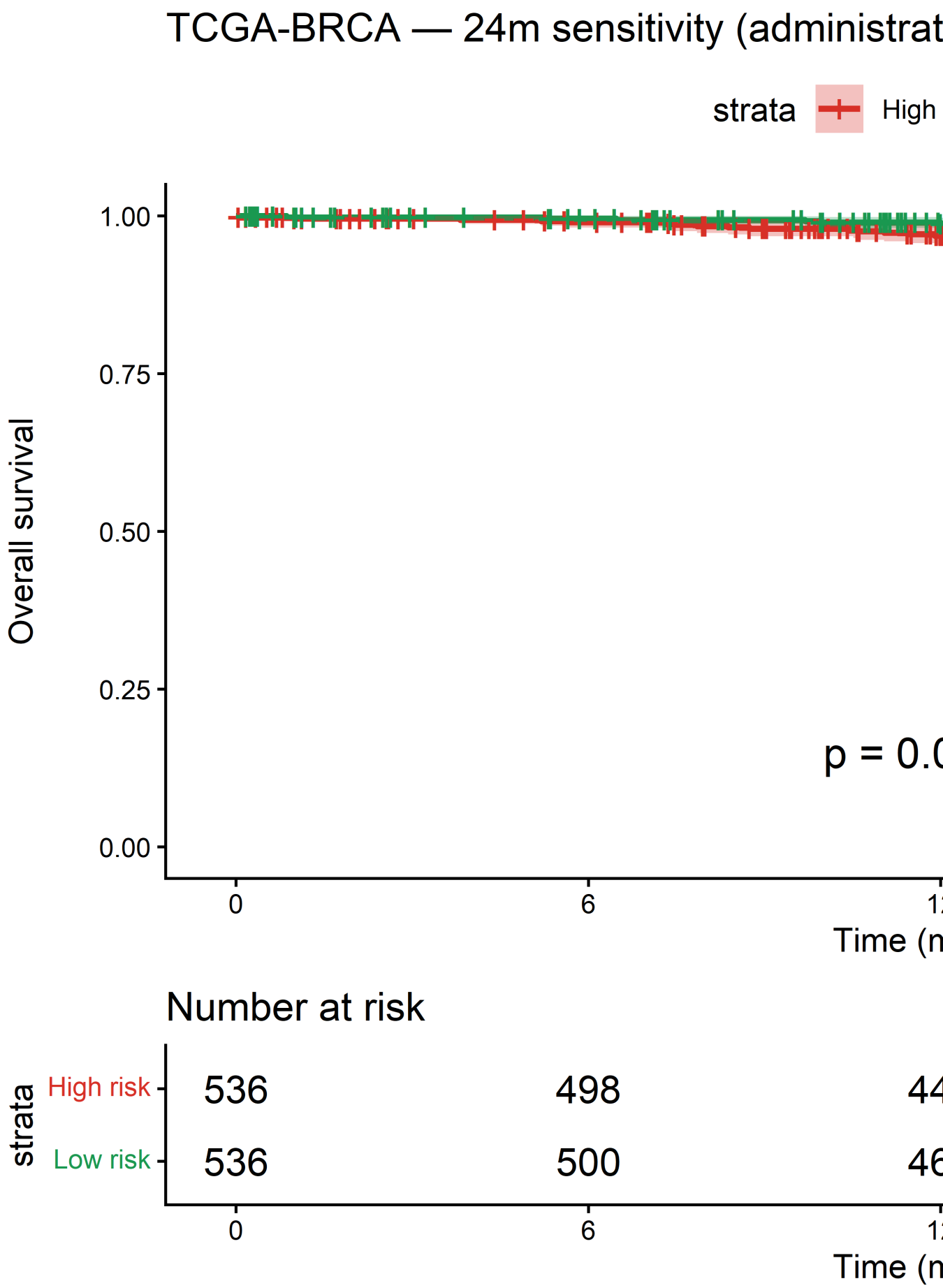

Figure 11: TCGA-BRCA 24-month sensitivity analysis. Patients censored at 24 months. HR=1.64 (95%CI 1.24-2.16), p=0.00047.

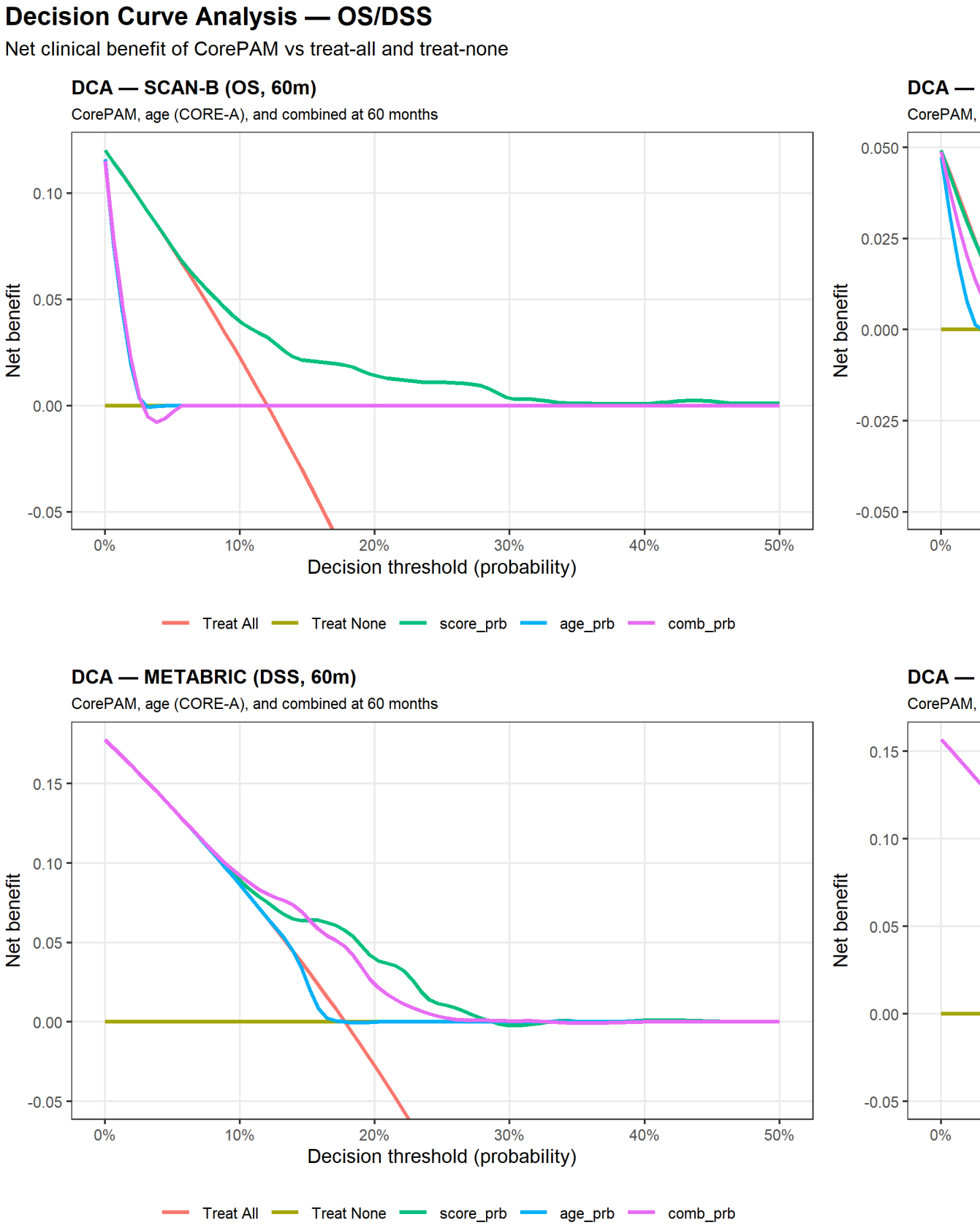

Figure 12: Decision curve analysis (OS). Net benefit of CorePAM score versus treat-all and treat-none strategies at clinically relevant probability thresholds. Horizons: 60 months (SCAN-B, METABRIC, GSE20685) and 24 months (TCGA-BRCA).

Supplementary Tables

Table 7: Expanded CORE-A analysis: CorePAM independence from anatomical staging. Multivariable Cox regression adjusting for age, ER status (where available), T-stage, and nodal status. CorePAM HR per 1 SD with 95% CI. Delta C-index = C(clinical+CorePAM) minus C(clinical alone) for the expanded model.

Cohort	N	Covariates	HR (95% CI)	p-value	C clinical	C clinical+CorePAM	Delta C
SCANB	2948	age+ER+T+N	1.79 (1.57–2.05)	2.22e-17	0.7657	0.7881	0.0224
TCGA_BRCA	1105	age+T+N	1.35 (1.15–1.58)	0.0002	0.7190	0.7455	0.0265
METABRIC	1883	age+ER+T+N	1.34 (1.23–1.46)	8.79e-11	0.6627	0.6883	0.0256
GSE20685	327	age+T+N	1.47 (1.15–1.86)	0.002	0.6972	0.7219	0.0247

Table 8: Head-to-head C-index comparison: CorePAM vs research-based ROR-S and OncotypeDX-RS. ROR-S computed from PAM50 centroid correlations (50 genes); ODX-RS from weighted gene expression sums (21 genes). All scores standardised (per 1 SD).

Cohort	N	PAM50 genes	CorePAM C	ROR-S C	ODX-RS C	CorePAM HR	ROR-S HR	ODX-RS HR
SCANB	3069	47	0.698	0.641	0.607	1.92	1.56	1.40
TCGA_BRCA	1072	46	0.624	0.602	0.586	1.20	1.25	1.13
METABRIC	1978	47	0.638	0.643	0.665	1.41	1.53	1.51
GSE20685	327	42	0.623	0.647	0.648	1.40	1.57	1.59

Declarations

Ethics approval and consent to participate: Not applicable. This study uses exclusively publicly available de-identified data (GEO, GDC, cBioPortal). No new human subjects research was conducted.

Consent for publication: Not applicable.

Competing interests: The author declares no competing interests.

Funding: No external funding was received. This study was conducted independently by the author.

Authors' contributions: RB designed the study, performed all analyses, and wrote the manuscript.

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Data availability: All raw data are available through GEO (GSE96058, GSE25066, GSE20194, GSE32646, GSE22226, GSE194040), GDC (TCGA-BRCA), and cBioPortal (METABRIC). Analysis code and processed results are available at https://github.com/RafaelBotan/Phd-Genomic/tree/main.

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